# MILITARY SPECIFICATION

## SWITCH ARM

### 1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation of delivery criteria for one type of switch arm used on the Atomic Demolition Charge.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8825429 - Switch Arm
8883534 - Bag Packing for Repair Parts
EL-8825429 - Equipment List

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switch arm shall comply with all requirements specified on the drawing (dwg) 8825429, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or ony other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanges process, in accordance with the same drawings, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.



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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved the the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II) and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.



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Table I

Major Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-72 | all | 140-141 | 124 | 256-259 | 177 | 541-549 | 230 |
| 73 | 72 | 142 | 125 | 260-262 | 178 | 550-559 | 231 |
| 74 | 73 | 143-144 | 126 | 263-265 | 179 | 560-569 | 232 |
| 75 | 74 | 145 | 127 | 266-268 | 180 | 570-579 | 233 |
| 76 | 75 | 146-147 | 128 | 269-271 | 181 | 580-590 | 234 |
| 77 | 76 | 148-149 | 129 | 272-275 | 182 | 591-601 | 235 |
| 78 | 77 | 150 | 130 | 276-278 | 183 | 602-613 | 236 |
| 79 | 78 | 151-152 | 131 | 279-282 | 184 | 614-625 | 237 |
| 80 | 79 | 153-154 | 132 | 283-285 | 185 | 626-637 | 238 |
| 81-82 | 80 | 155-156 | 133 | 286-289 | 186 | 638-649 | 239 |
| 83 | 81 | 157 | 134 | 290-293 | 187 | 650-662 | 240 |
| 84 | 82 | 158-159 | 135 | 294-296 | 188 | 663-676 | 241 |
| 85 | 83 | 160-161 | 136 | 297-300 | 189 | 677-690 | 242 |
| 86 | 84 | 162-163 | 137 | 301-304 | 190 | 691-704 | 243 |
| 87 | 85 | 164-165 | 138 | 305-308 | 191 | 705-719 | 244 |
| 88-89 | 86 | 166-167 | 139 | 309-312 | 192 | 720-735 | 245 |
| 90 | 87 | 168 | 140 | 313-316 | 193 | 736-751 | 246 |
| 91 | 88 | 169-170 | 141 | 317-320 | 194 | 752-768 | 247 |
| 92 | 89 | 171-172 | 142 | 321-324 | 195 | 769-785 | 248 |
| 93 | 90 | 173-174 | 143 | 325-329 | 196 | 786-804 | 249 |
| 94-95 | 91 | 175-176 | 144 | 330-333 | 197 | 805-823 | 250 |
| 96 | 92 | 177-178 | 145 | 334-338 | 198 | 824-842 | 251 |
| 97 | 93 | 179-180 | 146 | 339-342 | 199 | 843-863 | 252 |
| 98 | 94 | 181-182 | 147 | 343-347 | 200 | 864-884 | 253 |
| 99-100 | 95 | 183-184 | 148 | 348-352 | 201 | 885-907 | 254 |
| 101 | 96 | 185-187 | 149 | 353-356 | 202 | 908-930 | 255 |
| 102 | 97 | 188-189 | 150 | 357-361 | 203 | 931-955 | 256 |
| 103 | 98 | 190-191 | 151 | 362-366 | 204 | 956-981 | 257 |
| 104-105 | 99 | 192-193 | 152 | 367-372 | 205 | 982-1008 | 258 |
| 106 | 100 | 194-195 | 153 | 373-377 | 206 | 1009-1036 | 259 |
| 107 | 101 | 196-197 | 154 | 378-382 | 207 | 1037-1066 | 260 |
| 108-109 | 102 | 198-200 | 155 | 383-388 | 208 | 1067-1098 | 261 |
| 110 | 103 | 201-202 | 156 | 389-393 | 209 | 1099-1131 | 262 |
| 111 | 104 | 203-204 | 157 | 394-399 | 210 | 1132-1166 | 263 |
| 112-113 | 105 | 205-207 | 158 | 400-405 | 211 | 1167-1204 | 264 |
| 114 | 106 | 208-209 | 159 | 406-411 | 212 | 1205-1243 | 265 |
| 115 | 107 | 210-212 | 160 | 412-417 | 213 | 1244-1285 | 266 |
| 116-117 | 108 | 213-214 | 161 | 418-424 | 214 | 1286-1330 | 267 |
| 118 | 109 | 215-217 | 162 | 425-430 | 215 | 1331-1378 | 268 |
| 119 | 110 | 218-219 | 163 | 431-437 | 216 | 1379-1429 | 269 |
| 120-121 | 111 | 220-222 | 164 | 438-443 | 217 | 1430-1483 | 270 |
| 122 | 112 | 223-224 | 165 | 444-450 | 218 | 1484-1542 | 271 |
| 123-124 | 113 | 225-227 | 166 | 451-457 | 219 | 1543-1605 | 272 |
| 125 | 114 | 228-230 | 167 | 458-465 | 220 | 1606-1673 | 273 |
| 126-127 | 115 | 231-232 | 168 | 466-472 | 221 | 1674-1747 | 274 |
| 128 | 116 | 233-235 | 169 | 473-480 | 222 | 1748-1827 | 275 |
| 129-130 | 117 | 236-238 | 170 | 481-488 | 223 | 1828-1914 | 276 |
| 131 | 118 | 239-241 | 171 | 489-496 | 224 | 1915-2010 | 277 |
| 132-133 | 119 | 242-243 | 172 | 497-504 | 225 | 2011-2115 | 278 |
| 134 | 120 | 244-246 | 173 | 505-513 | 226 | 2116-2232 | 279 |
| 135-136 | 121 | 247-249 | 174 | 514-521 | 227 | 2233-2361 | 280 |
| 137 | 122 | 250-252 | 175 | 522-530 | 228 | over 2361 | 298 |
| 138-139 | 123 | 253-255 | 176 | 531-540 | 229 | | |

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Table II
Minor Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-16 | all | 28-29 | 23 | 52-56 | 31 | 132-156 | 39 |
| 17 | 16 | 30-31 | 24 | 57-62 | 32 | 157-191 | 40 |
| 18 | 17 | 32-34 | 25 | 63-69 | 33 | 192-244 | 41 |
| 19-20 | 18 | 35-36 | 26 | 70-76 | 34 | 245-334 | 42 |
| 21 | 19 | 37-40 | 27 | 77-86 | 35 | 335-519 | 43 |
| 22-23 | 20 | 41-43 | 28 | 87-97 | 36 | 520-1120 | 44 |
| 24-25 | 21 | 44-47 | 29 | 98-112 | 37 | over 1120 | 45 |
| 26-27 | 22 | 48-51 | 30 | 113-131 | 38 | | |

4.3.2.1 Classification of characteristics (see 3.2). Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch arm (see dwg 8825429).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Thickness (.062) | Gage |
| | 102. Diameter of hole, max (.136) | Gage |
| | 103. Contour of switch arm and slot in relation to hole | Gage |
| | 104. Material (see 3.1) | Visual |
| Minor: | | |
| | 201. Workmanship (see 3.3) | Visual/Manual |

4.3.2.1.2 Preparation for delivery (see 5.1).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Preservation, packaging, packing and marking in accordance with MIL-STD-1169 | Visual/Manual |
| Minor: | None defined. | |



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4.3.2.1.3 Bag, packing for repair parts (see dwg 8883534 covering drawing part number 8883534-1 and 5.2).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Material (see 3.1) | Visual |
| | 102. Length before packing, min | Measure* |
| | 103. Width before packing, min | Messure* |
| | 104. Correctly packed | Visual/Manual |
| | 105. Proper heat seal | Visual/Manual |
| | 106. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116 | Visual |
| | 107. No crack or hole in bag | Visual |

*Measure with suitable scale.

| | | |
|---|---|---|
| Minor: | | |
| | 201. Marking | Visual |
| | 202. Workmanship (see 3.3) | Visual/Manual |

4.3.2.1.4 Packing and marking for storage and shipments (see dwg 8808362 and 5.2).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Gross load weight, max | Measure* |
| | 102. Contents tight in package | Manual |
| | 103. Correctly packed | Visual/Manual |
| | 104. Material (see 3.1) | Visual |

*Determine weight with a suitable scale.

| | | |
|---|---|---|
| Minor: | | |
| | 201. Marking | Visual |
| | 202. Workmanship (see 3.3) | Visual/Manual |

4.3.3 Testing. Not applicable.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration



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instructions specified on the Equipment Lists (EL) (see EL-8825429). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. Not applicable.

## 5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The switch arm shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The switch arm shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The switch arm shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The switch arm shall be packaged in accordance with dwg 8883534 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The switch arm shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The switch arm shall be packed in accordance with dwgs 8883534 and 8808362.

5.2.3 Marking. Marking shall be as specified is accordance with dwgs 8883534 and 8808362 as applicable.



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or Repair Part.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1667 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1135-A124



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-50231(MU) SWITCH ARM

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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